

String Manipulation Operations

Q. What is a String in Data Structures? Explain the common string manipulation operations with examples.

****Definition to Remember****

In C, a
refers to
for all m

1. String Representation in Memory

```
char str[] = "HELLO";
```

Memory

2. Common String Manipulation Operations

Function	Operation	Description
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:--- :-

3. Code Example

```
#include <stdio.h>
```

```
#include
```

```
int main() {
```

```
char str
```

```
// 1. Length
```

```
printf("L
```

```
// 2. Copy
```

```
strcpy(c
```

```
// 3. Concatenation
```

```
strcat(d
```

```
// 4. Comparison
```

```
return 0;
```

```
---
```

*****Must-Write Points (for 10 marks)*****

```
---
```

*****Quick Recall*****

```
---
```

Custom String Manipulation Algorithms & C Codes

1. Custom String Operations: `strlen`, `strcpy`, `strcat`, `strrev` (C Code)

```
#include <stdio.h>
```

```
// Custom strlen
```

```
// Custom strcpy
```

```
// Custom strcat
```

```
// Custom String Reverse
```

```
int main() {
```

```
printf("Length of s1: %d\n", my_strlen(s1));
```

```
my_strcat(s1, s2);
```

```
my_strrev(s1);
```

```
return 0;
```

```
---
```

2. Naive Pattern Matching Algorithm (C Code)

```
#include <stdio.h>
```

```
void naivePatternSearch(char text[], char pattern[]) {
```

```
for (int i = 0; i <= n - m; i++) {
```

```
int main() {
```

```
void my
```

```
char s1
```

```
printf("C
```

```
printf("F
```

```
}
```

```
#include
```

```
int n = s
```

```
int j;
```

```
char tex
```

```
printf("Text: %s\nPattern: %s\n\n", text, pattern);
```

naivePa

```
return 0;
```

```
}
```